PROBLEM 35

String with Two Particles on a Cylinder

Original statement · Problems.pdf, p. 23

Multi-Particle Cylinder Contact

Two heavy particles P and Q of mass m each are connected by an inextensible massless string of constant length L. Part of this string rests on top of a cylinder of radius R and center at O. The string can slide over the cylinder without friction, and remains tense at all times. Furthermore, P and Q are also connected by a spring of zero natural length and stiffness k. We define the inertial reference frame S_0 as shown in the figure. We denote as A and B the points of tangency of the string with the cylinder.

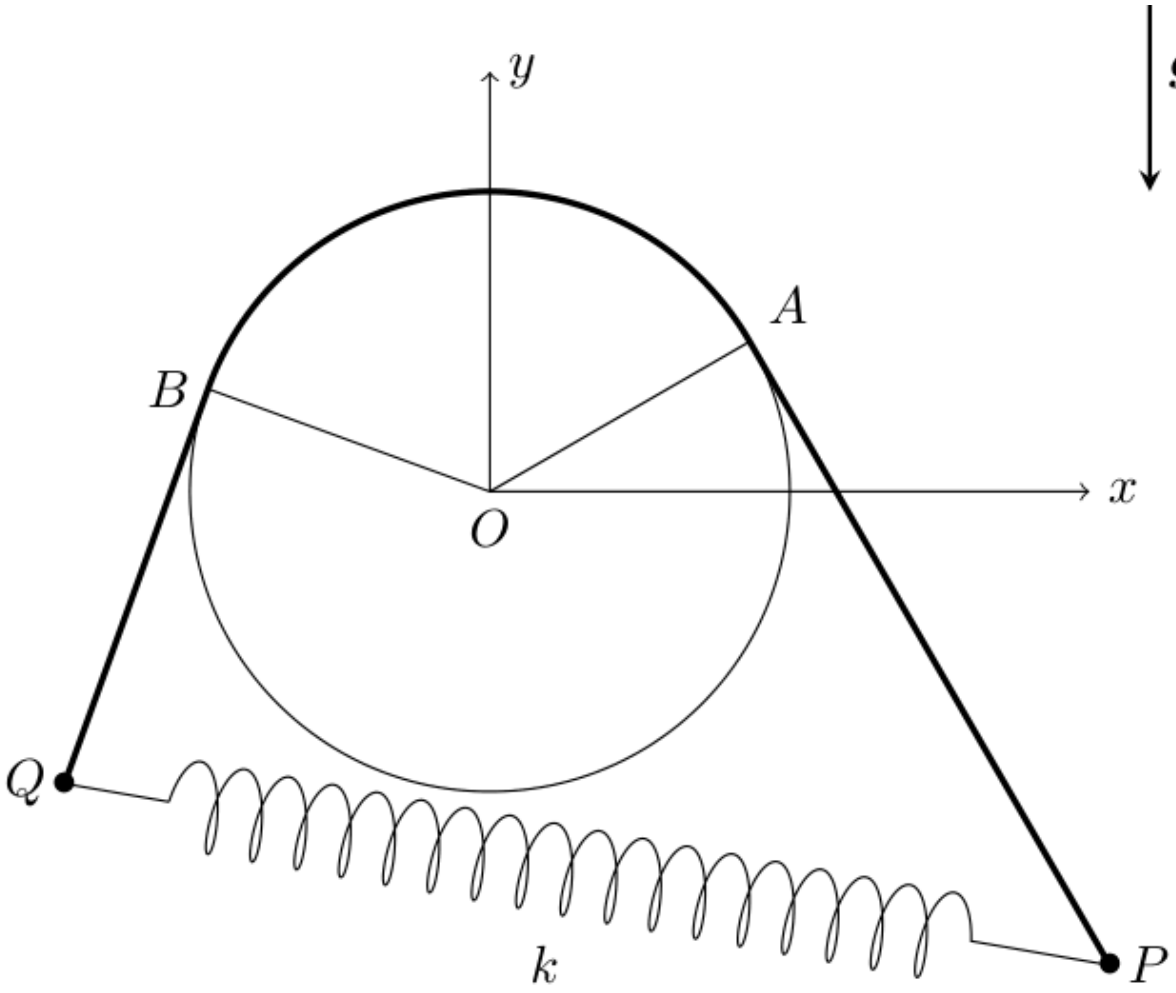


Figure: String over the cylinder of radius R centred at O with tangency points A and B, particles P and Q joined by the spring k (Problems.pdf, p. 23).

(a) How many degrees of freedom does this system have? Define your own generalized coordinates to parametrize the configuration of the system. You will use them in the rest of the problem.

(b) Write down the general expression of the position, velocity and acceleration vectors of P and Q relative to the S_0 reference frame as a function of your chosen coordinates and their time derivatives.

(c) Write down the expression of all forces acting on P and Q.

(d) Discuss whether the mechanical energy of P is conserved; if so, write down the expression of the corresponding potential energy.

(e) Consider now that we restrict the motion of the system to symmetric configurations about the Oy axis only. What is the number of degrees of freedom now? Find the mathematical relations that exist between your previously-defined coordinates in this case.

(f) Find the approximate value of your coordinates when the system is in equilibrium and the configuration is symmetric about the Oy axis. To do so, assume that the deviation from the vertical of the AP and BQ segments is small, and linearize the equations of motion. Find the angular frequency of the oscillations around this equilibrium configuration.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** two heavy particles P and Q of mass m are joined by an inextensible massless string of length L that rests on a fixed smooth cylinder of radius R and centre O, and by a spring of zero natural length and stiffness k. The motion is planar, in Oxy with Oy upward. A (right) and B (left) are the tangency points of the string with the cylinder. The string stays taut and the spring does not touch the cylinder.
- **Constraints:** two free particles in the plane have 4 coordinates. The inextensible string, taut and in contact with the cylinder, gives one holonomic constraint (unilateral: $T \geq 0$), so $\mathrm{CDOF} = 4 - 1 = 3$. The smooth cylinder acts on the string, not on the particles.
- **Coordinates:** A is at the polar angle α from Ox and B at the polar angle β. The straight segments AP and BQ have lengths ξ_P and ξ_Q, and the string wraps the arc AB of length $R(\beta - \alpha)$ on the cylinder:

$$\xi_P + \xi_Q + R(\beta - \alpha) = L$$

The generalised coordinates are (α, β, ξ_P), with $\xi_Q = L - \xi_P - R(\beta - \alpha)$ (the four quantities $\alpha, \beta, \xi_P, \xi_Q$ are used below together with this one relation).

- **Parameters:** m [kg], L [m], R [m], k [N/m], g [m/s^2]; $\ell_0 = (L - \pi R)/2$ [m] is defined in (e).

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $S_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ fixed to the cylinder.
- **Bases following the contact points:** $\mathbf{n}$ is the outward normal of the cylinder and $\mathbf{t}$ the unit vector along the straight string, from the tangency point to the particle (clockwise at A, counterclockwise at B):

$$\mathbf{n}_A = \cos\alpha\,\mathbf{i} + \sin\alpha\,\mathbf{j}, \quad \mathbf{t}_A = \sin\alpha\,\mathbf{i} - \cos\alpha\,\mathbf{j}, \qquad \mathbf{n}_B = \cos\beta\,\mathbf{i} + \sin\beta\,\mathbf{j}, \quad \mathbf{t}_B = -\sin\beta\,\mathbf{i} + \cos\beta\,\mathbf{j}$$

- **Changes of basis** (columns are the components of $\mathbf{t}_A, \mathbf{n}_A, \mathbf{k}$ and of $\mathbf{n}_B, \mathbf{t}_B, \mathbf{k}$):

$$[{}_0R_A] = \begin{pmatrix} \sin\alpha & \cos\alpha & 0 \\ -\cos\alpha & \sin\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad [{}_0R_B] = \begin{pmatrix} \cos\beta & -\sin\beta & 0 \\ \sin\beta & \cos\beta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\det[{}_0R_A] = \sin^2\alpha + \cos^2\alpha = 1, \quad \det[{}_0R_B] = \cos^2\beta + \sin^2\beta = 1, \quad [{}_0R_A][{}_0R_A]^T = [{}_0R_B][{}_0R_B]^T = I$$

Multiplying out, RR^T has diagonal entries $\cos^2 + \sin^2 = 1$ of the rotation angle (and 1) and off-diagonal entries $\pm(\cos\sin - \sin\cos) = 0$, so it is the identity.

- **Time derivatives** (each basis turns about $\mathbf{k}$ at the rate of its angle):

$$\dot{\mathbf{t}}_A = \dot\alpha\,\mathbf{n}_A, \quad \dot{\mathbf{n}}_A = -\dot\alpha\,\mathbf{t}_A, \qquad \dot{\mathbf{n}}_B = \dot\beta\,\mathbf{t}_B, \quad \dot{\mathbf{t}}_B = -\dot\beta\,\mathbf{n}_B$$

PHASE 3

Derivation

3.1 Kinematics

(b) Position, velocity and acceleration

Pedagogical Justification: each particle is at the tangency point plus the straight segment, $\mathbf{r}_P = R\,\mathbf{n}_A + \xi_P\,\mathbf{t}_A$ and $\mathbf{r}_Q = R\,\mathbf{n}_B + \xi_Q\,\mathbf{t}_B$. The derivatives use the product rule and the formulas of Phase 2:

$$\mathbf{v}_P = -R\dot\alpha\,\mathbf{t}_A + \dot\xi_P\,\mathbf{t}_A + \xi_P\dot\alpha\,\mathbf{n}_A = \left(\dot\xi_P - R\dot\alpha\right)\mathbf{t}_A + \xi_P\dot\alpha\,\mathbf{n}_A$$

$$\mathbf{a}_P = \left(\ddot\xi_P - R\ddot\alpha\right)\mathbf{t}_A + \left(\dot\xi_P - R\dot\alpha\right)\dot\alpha\,\mathbf{n}_A + \left(\dot\xi_P\dot\alpha + \xi_P\ddot\alpha\right)\mathbf{n}_A - \xi_P\dot\alpha^2\,\mathbf{t}_A = \left(\ddot\xi_P - R\ddot\alpha - \xi_P\dot\alpha^2\right)\mathbf{t}_A + \left(\xi_P\ddot\alpha + 2\dot\xi_P\dot\alpha - R\dot\alpha^2\right)\mathbf{n}_A$$

$$\mathbf{v}_Q = \left(\dot\xi_Q + R\dot\beta\right)\mathbf{t}_B - \xi_Q\dot\beta\,\mathbf{n}_B, \qquad \mathbf{a}_Q = \left(\ddot\xi_Q + R\ddot\beta - \xi_Q\dot\beta^2\right)\mathbf{t}_B - \left(\xi_Q\ddot\beta + 2\dot\xi_Q\dot\beta + R\dot\beta^2\right)\mathbf{n}_B$$

$\mathbf{a}_Q$ follows from $\mathbf{v}_Q$ exactly as $\mathbf{a}_P$ from $\mathbf{v}_P$, with $\dot{\mathbf{t}}_B$ and $\dot{\mathbf{n}}_B$ of Phase 2 and β in place of α. In terms of the chosen coordinates (α, β, ξ_P): $\xi_Q = L - \xi_P - R(\beta - \alpha)$, $\dot\xi_Q = -\dot\xi_P - R(\dot\beta - \dot\alpha)$ (so $\dot\xi_Q + R\dot\beta = R\dot\alpha - \dot\xi_P$) and $\ddot\xi_Q + R\ddot\beta = R\ddot\alpha - \ddot\xi_P$.

3.2 Forces

(c) Forces on P and Q

- **Weights:** $-mg\,\mathbf{j}$ on each particle.
- **String tension** $T \geq 0$ (the same on both sides: the cylinder is smooth and the string massless): $-T\,\mathbf{t}_A$ on P and $-T\,\mathbf{t}_B$ on Q, directed toward the tangency points.
- **Spring** (zero natural length): $\mathbf{F}_{s,P} = k\,\mathbf{d}$ and $\mathbf{F}_{s,Q} = -k\,\mathbf{d}$, with $\mathbf{d} = \mathbf{r}_Q - \mathbf{r}_P$.
- **Inertia forces:** none. The projections of the weight on the bases are $\mathbf{j}\cdot\mathbf{t}_A = -\cos\alpha, \mathbf{j}\cdot\mathbf{n}_A = \sin\alpha, \mathbf{j}\cdot\mathbf{t}_B = \cos\beta, \mathbf{j}\cdot\mathbf{n}_B = \sin\beta$.

3.3 Equations of Motion

Pedagogical Justification: Newton's second law for P and Q, projected on the bases of the contact points, gives four scalar equations; together with the length constraint there are 5 equations for the 5 unknowns $\alpha, \beta, \xi_P, \xi_Q, T$:

$$m\left(\ddot{\xi}_P - R\ddot{\alpha} - \xi_P\dot{\alpha}^2\right) = -T + mg\cos\alpha + k\,\mathbf{d}\cdot\mathbf{t}_A$$

$$m\left(\xi_P\ddot{\alpha} + 2\dot{\xi}_P\dot{\alpha} - R\dot{\alpha}^2\right) = -mg\sin\alpha + k\,\mathbf{d}\cdot\mathbf{n}_A$$

$$m\left(\ddot{\xi}_Q + R\ddot{\beta} - \xi_Q\dot{\beta}^2\right) = -T - mg\cos\beta - k\,\mathbf{d}\cdot\mathbf{t}_B$$

$$-m\left(\xi_Q\ddot{\beta} + 2\dot{\xi}_Q\dot{\beta} + R\dot{\beta}^2\right) = -mg\sin\beta - k\,\mathbf{d}\cdot\mathbf{n}_B$$

- **Decoupling.** Subtracting the first and third equations removes the tension T; the remaining equations, with the length constraint, determine the three coordinates, and T is computed afterwards.

3.4 Integration and Answers

(a) Degrees of freedom

- $\mathrm{CDOF} = 3$ (see Phase 1): for example (α, β, ξ_P).

(d) Mechanical energy

Pedagogical Justification: the energy of P alone is not conserved: the tension and the spring both do work on it, and that work is exchanged with Q. For the system $P + Q$, the only forces that could dissipate energy are the tensions, so their total power is computed first, with $\mathbf{t}_A \cdot \mathbf{v}_P = \dot{\xi}_P - R\dot{\alpha}$ and $\mathbf{t}_B \cdot \mathbf{v}_Q = \dot{\xi}_Q + R\dot{\beta}$:

$$\dot{W}_T = -T\,\mathbf{t}_A \cdot \mathbf{v}_P - T\,\mathbf{t}_B \cdot \mathbf{v}_Q = -T\,\frac{d}{dt}\left[\xi_P + \xi_Q + R(\beta - \alpha)\right] = -T\dot{L} = 0$$

- The tensions do zero net work. Gravity and the spring are conservative, so the total mechanical energy of the system is conserved, with the potential energy

$$V = mg\,(y_P + y_Q) + \tfrac{1}{2}k\,|\mathbf{r}_Q - \mathbf{r}_P|^2\,, \qquad E = \tfrac{1}{2}m\left(v_P^2 + v_Q^2\right) + V = \text{const}$$

(e) Symmetric configurations about Oy

> **Pedagogical Justification:** if the motion keeps the mirror symmetry about Oy, then $\beta = \pi - \alpha$ and $\xi_P = \xi_Q = \xi$. The length constraint becomes $2\xi + R(\pi - 2\alpha) = L$, i.e.

$$\xi(\alpha) = \ell_0 + R\alpha, \qquad \ell_0 = \frac{L - \pi R}{2}$$

- The system has $\mathrm{CDOF} = 1$, the angle α. The relations between the previous coordinates are $\beta = \pi - \alpha$, $\xi_P = \xi_Q = \ell_0 + R\alpha$ and $\dot{\xi} = R\dot{\alpha}$.

(f) Equilibrium and small oscillations

> **Pedagogical Justification:** with the symmetry, one coordinate α describes the system. The coordinates of P are $x_P = R\cos\alpha + \xi\sin\alpha$ and $y_P = R\sin\alpha - \xi\cos\alpha$. Differentiating with respect to α (with $\xi' = R$):

$$x_P' = -R\sin\alpha + R\sin\alpha + \xi\cos\alpha = \xi\cos\alpha, \qquad y_P' = R\cos\alpha - R\cos\alpha + \xi\sin\alpha = \xi\sin\alpha$$

so $v_P = \xi\dot{\alpha}$. By symmetry $x_Q = -x_P$, $y_Q = y_P$ and $|\mathbf{r}_Q - \mathbf{r}_P| = 2x_P$:

$$K = 2\cdot\tfrac{1}{2}m\xi^2\dot{\alpha}^2 = m\xi^2\dot{\alpha}^2, \qquad V(\alpha) = 2mgy_P + \tfrac{1}{2}k(2x_P)^2 = 2mgy_P + 2kx_P^2$$

$$V'(\alpha) = 2mg\xi\sin\alpha + 4kx_P\xi\cos\alpha = 2\xi f(\alpha), \qquad f(\alpha) = mg\sin\alpha + 2k\cos\alpha\,(R\cos\alpha + \xi\sin\alpha)$$

- **Equilibrium.** $V'(\alpha_0) = 0$, i.e. $f(\alpha_0) = 0$ (since $\xi > 0$). Physically, the horizontal pull $2kx_P$ of the spring must be balanced by the horizontal component of the tension, which needs $\alpha_0 < 0$: the segments AP and BQ are slightly inclined inward. For a small deviation from the vertical, $|\alpha_0| \ll 1$, keep first-order terms ($\sin\alpha \approx \alpha$, $\cos\alpha \approx 1$, $\xi \approx \ell_0$):

$$mg\alpha_0 + 2k\,(R + \ell_0\alpha_0) = 0 \implies \alpha_0 \approx -\frac{2kR}{mg + 2k\ell_0}$$

and the other coordinates at equilibrium follow from the symmetry relations: $\beta_0 = \pi - \alpha_0 \approx \pi + \dfrac{2kR}{mg + 2k\ell_0}$ and $\xi_0 = \ell_0 + R\alpha_0 \approx \ell_0 - \dfrac{2kR^2}{mg + 2k\ell_0}$.

- **Equation of motion.** From $\frac{d}{dt}(K + V) = 0$ with the chain rule and $\dot\xi = R\dot\alpha$:

$$2m\xi^2\dot\alpha\ddot\alpha + 2m\xi R\dot\alpha^3 + 2\xi f\dot\alpha = 0 \implies m\xi\ddot\alpha + mR\dot\alpha^2 + f(\alpha) = 0$$

This is the $\mathbf{n}_A$ equation of 3.3 with $\dot\xi_P = R\dot\alpha$ and $\mathbf{d}\cdot\mathbf{n}_A = -2x_P\cos\alpha$.

- **Linearisation.** Put $\alpha = \alpha_0 + \delta$ with δ small: $\dot\alpha^2$ is second order and $f(\alpha) \approx f'(\alpha_0)\delta$, with $f'(\alpha) = mg\cos\alpha + 2k\left[\xi\cos^2\alpha - x_P\sin\alpha\right] \approx mg + 2k\ell_0$:

$$\ddot\delta + \frac{mg + 2k\ell_0}{m\ell_0}\,\delta = 0 \implies \omega^2 = \frac{g}{\ell_0} + \frac{2k}{m}, \qquad \boxed{\omega = \sqrt{\frac{g}{\ell_0} + \frac{2k}{m}}}$$

PHASE 4

Interpretation & Checks

- **Limit $k \to 0$:** $\omega = \sqrt{g/\ell_0}$, the pendulum of length ℓ_0 (the free length of each hanging segment).

- **Limit $g \to 0$:** $\omega = \sqrt{2k/m}$, the frequency of two particles of mass m joined by a spring k (reduced mass $m/2$ and stiffness k). In this limit $\alpha_0 \to -R/\ell_0$, which is small only if $R \ll \ell_0$; otherwise the limit lies outside the small-angle hypothesis.
- **Dimensional check:** $[g/\ell_0] = \mathrm{s}^{-2}$, $[2k/m] = \mathrm{N\ m^{-1} kg^{-1}} = \mathrm{s}^{-2}$; $[\alpha_0] = [kR/(mg + 2k\ell_0)] = 1$.
- **Check against the official key:** `Problems.pdf` gives only the spring force, $\mathbf{F}_{S,P} = -\mathbf{F}_{S,Q} = k\overrightarrow{PQ} = k(\mathbf{r}_Q - \mathbf{r}_P)$ written in its own coordinates, which is the same vector as $k\,\mathbf{d}$ above. The remaining results are verified by independent derivation and numerically ($R = 1$ m, $\ell_0 = 2$ m, $m = 1$ kg, $k = 0.3$ N/m, $g = 9.81$ m/s^2): the exact root of $f(\alpha_0) = 0$ is $\alpha_0 = -0.054532$ rad against -0.054496 rad from the linear formula, and integrating $m\xi\ddot{\alpha} + mR\dot{\alpha}^2 + f = 0$ gives $\omega^2 = 5.6482$ s^{-2}, equal to $f'(\alpha_0)/(m\xi(\alpha_0))$; the difference with $g/\ell_0 + 2k/m = 5.505$ s^{-2} is the first-order correction $\xi(\alpha_0) = \ell_0 + R\alpha_0$ in place of ℓ_0.

Source: Problems.pdf